

Module 6: Computing (Database Services)

At the end of this module, you will be able to:

- **Differentiate** the types of managed database services offered by Amazon Web Services (AWS) and identify their recommended use.
- **Identify** some of the factors for consideration when selecting a database (engine and workloads).
- **Explain** the purpose and function of Amazon Relational Database Service (Amazon RDS), Amazon Aurora, and Amazon DynamoDB and related benefits of each.
- **Describe** the main features and benefits of Amazon Relational Database Service (Amazon RDS), Amazon Aurora, and Amazon DynamoDB.
- **Explain** the benefits of the AWS Database Migration Service (AWS DMS) and the capabilities of the AWS Schema Conversion Tool (AWS SCT).

Topics:

- AWS Database Services
- Amazon Relational Database Service (Amazon RDS)
- Amazon Aurora
- Amazon DynamoDB
- AWS Database Migration Service (AWS DMS)

Knowledge check

Instructor Demo:

- Create an Amazon RDS instance (Optional)

Lab: None

Exercise/Project:

- Hands-on Activity 6: Migrate database to Amazon RDS
- Project: Troubleshooting knowledge base

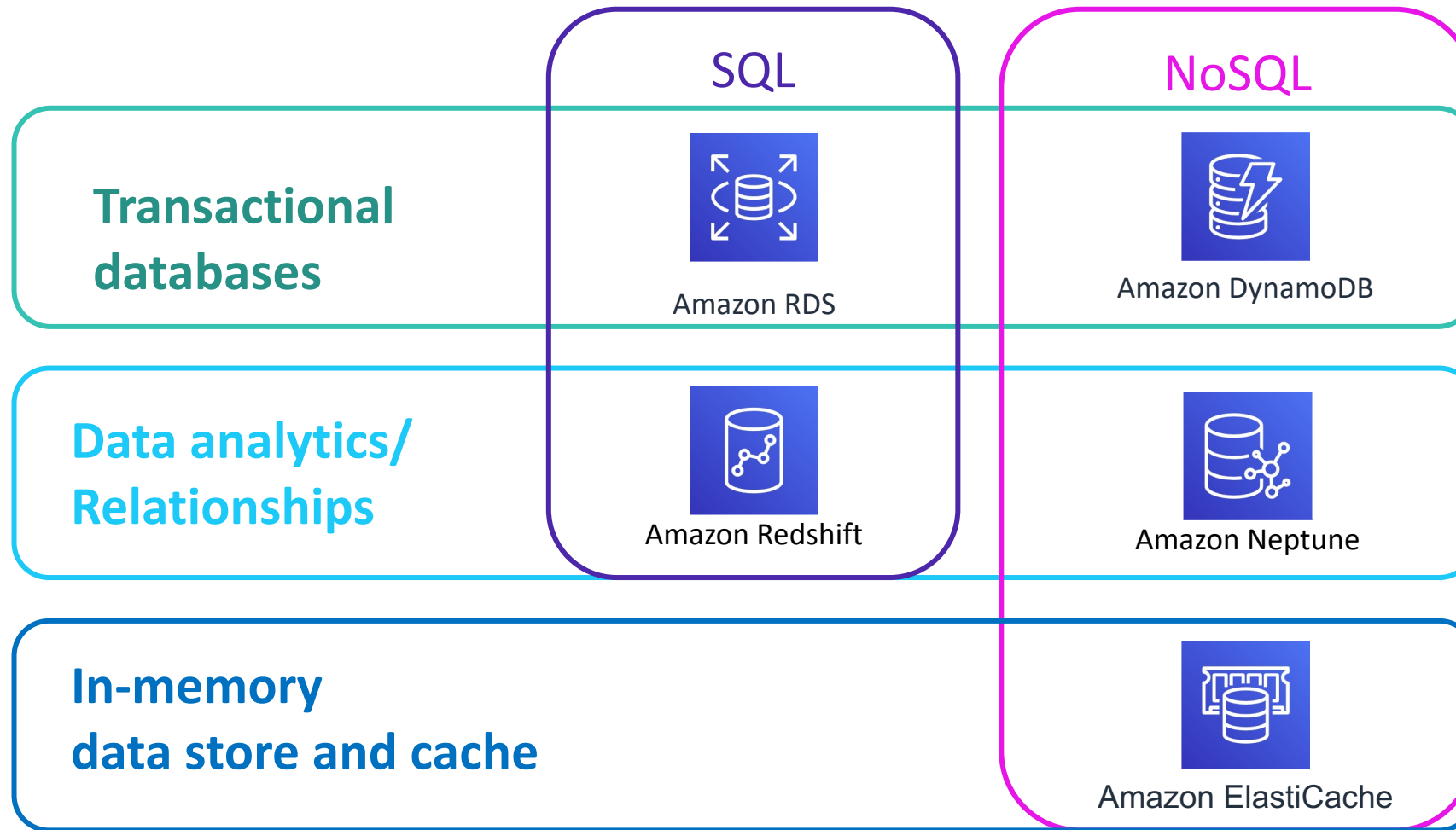
Troubleshooting Knowledge Base – Module 6 Focus

- Have your copy of the troubleshooting knowledge base template open for editing
- **Create new entries** as you encounter troubleshooting situations
- **Categories of focus** for this module:
 - Storage & Data Management

Module 6: Computing (Database Services)

AWS Database Services

AWS database options



SQL and NoSQL databases

	SQL	NoSQL
Data Storage	Rows and columns	Key-value
Schemas	Fixed	Dynamic
Querying	Uses SQL	Focuses on collection of documents
Scalability	Vertical	Horizontal

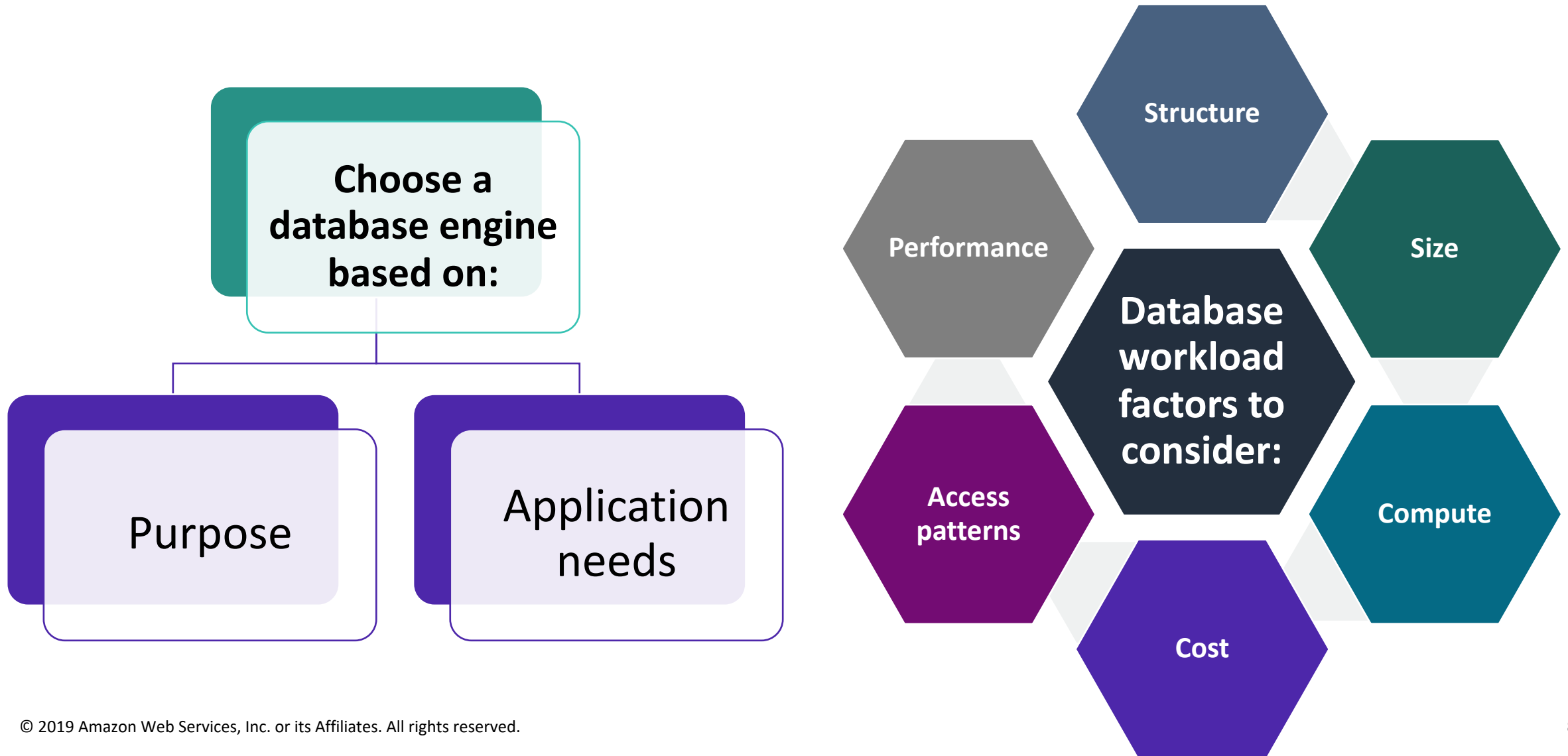
SQL

ISBN	Title	Author	Format
9182932465265	Cloud Computing Concepts	Wilson, Joe	Paperback
3142536475869	The Database Guru	Gomez, Maria	Ebook

NoSQL

```
{  
  "ISBN": 9182932465265,  
  "Title": "Cloud Computing Concepts",  
  "Author": "Wilson, Joe",  
  "Format": "Paperback"  
}
```

Choosing a database service



AWS database recommendations

If You Need	Product Type	Consider Using
Managed relational database that is launched with a choice of six popular database engines	Relational database	Amazon RDS
MySQL- and PostgreSQL-compatible relational database built for the cloud	Relational database	Amazon Aurora
Fully managed NoSQL database	NoSQL database	Amazon DynamoDB
Managed graph database	NoSQL database	Amazon Neptune
Managed Redis key-value store	NoSQL database	Amazon ElastiCache
Columnar storage SQL data warehouse	Relational database	Amazon Redshift

AWS database usage scenarios

Database Name	Best Used For
Amazon RDS Amazon Aurora	• Transactional applications like ERP, CRM, and ecommerce to log transactions and store structured data. • Amazon Aurora differentiator: Open source, high performance, and low cost
Amazon Redshift	• Analytic applications for operational reporting and querying terabyte- to exabyte-scale data.
Amazon ElastiCache	• Real time application use cases that require submillisecond latency, like gaming leaderboards, chat or messaging, streaming, and IoT.
Amazon Neptune	• Applications with use cases that require the navigation of highly connected data like social news feeds, recommendations, and fraud detection.
Amazon DynamoDB	• Internet-scale applications like hospitality, dating, and ridesharing that need to serve content and store structured and unstructured data.

Key Takeaways



- Managed databases simplify and automate many operational and management tasks.
- The choice of AWS managed database to use is based on application requirements.
- Each database offering has a primary use case -- make sure you are familiar with it.
- Unsupported database engines or database versions can be hosted on Amazon EC2.

Module 6: Computing (Database Services)

Amazon Relational Database Service (Amazon RDS)

Amazon RDS DB instances



Amazon RDS

DB Instance Class

CPU

Memory

Network Performance

DB Instance Storage

Magnetic

General Purpose (SSD)

Provisioned IOPS

DB Engines:

- Amazon Aurora
- MySQL
- SQL Server
- PostgreSQL
- MariaDB
- Oracle

Amazon RDS backup options include:

Automatic:

- Creates automated backups (full daily snapshot and transaction logs) of DB instance during the backup window

Manual:

- Creates a storage volume snapshot of your DB instance

Create snapshots: Command

To create a snapshot, the command is:

```
$ aws rds create-db-snapshot --db-instance-identifier  
mytestdb --db-snapshot-identifier mydbsnapshot
```

Restore DB from snapshot: Command

To restore a DB from a snapshot, the command is:

```
$ aws rds restore-db-instance-from-db-snapshot --db-instance-identifier mytestdb-new --db-snapshot-identifier mydbsnapshot
```

Copy RDS Snapshot: Command

To copy an RDS snapshot, the command is:

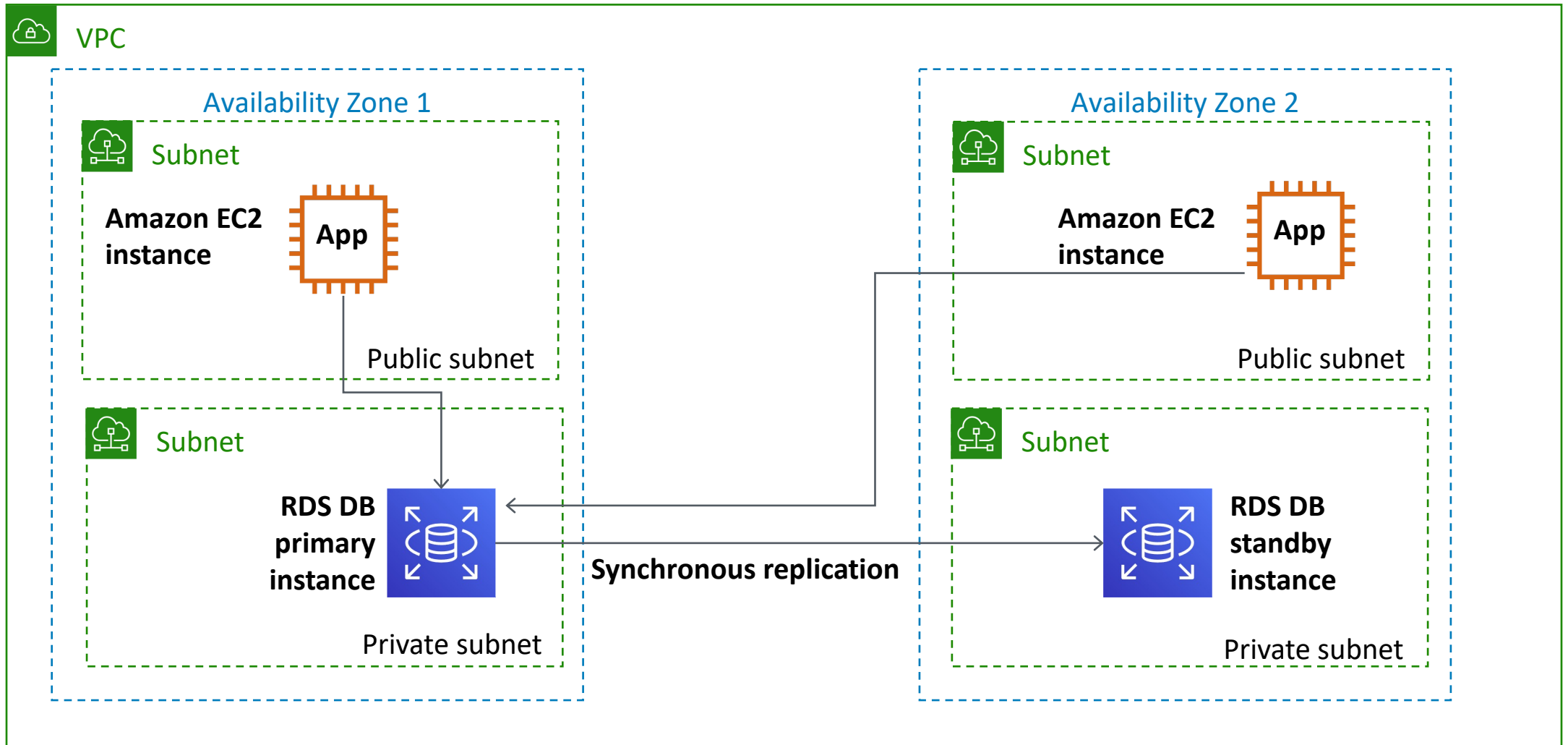
```
$ aws rds copy-db-snapshot --source-db-snapshot-identifier  
mydbsnapshot --target-db-snapshot-identifier mydbsnapshot-  
copy --copy-tags
```

Deleting a snapshot: Command

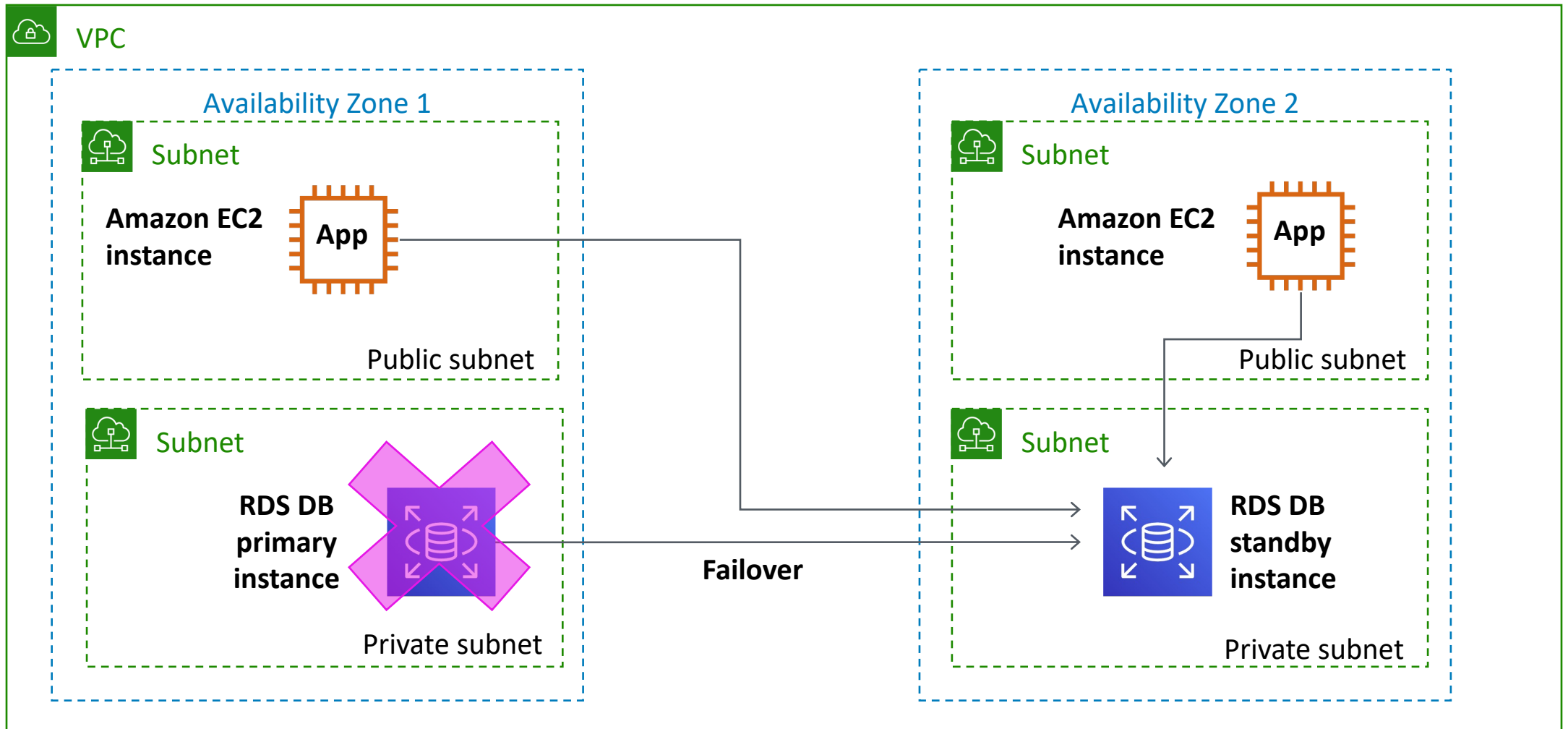
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```
$ aws rds delete-db-snapshot --db-snapshot-identifier  
mydbsnapshot
```

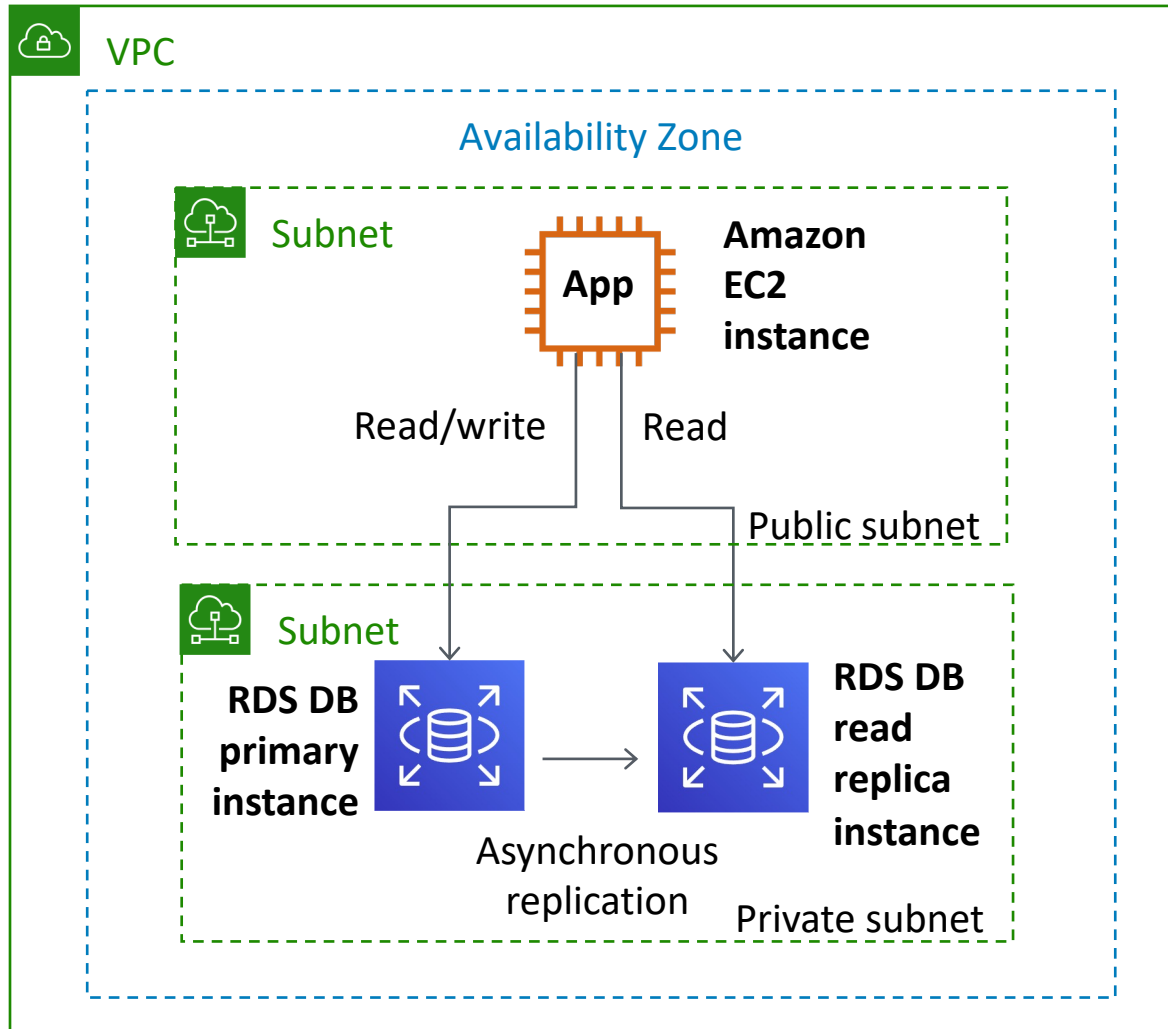
High availability with multi-AZ deployment: Replication



High availability with multi-AZ deployment: Failover



Amazon RDS scaling



Amazon RDS scaling:

- Instance class
 - Change instance class to increase computation and memory capacity
- Storage capacity
 - Scale storage for existing instances
- Read workloads could utilize read replicas

Optional instructor demo: Create an RDS instance

Key Takeaways

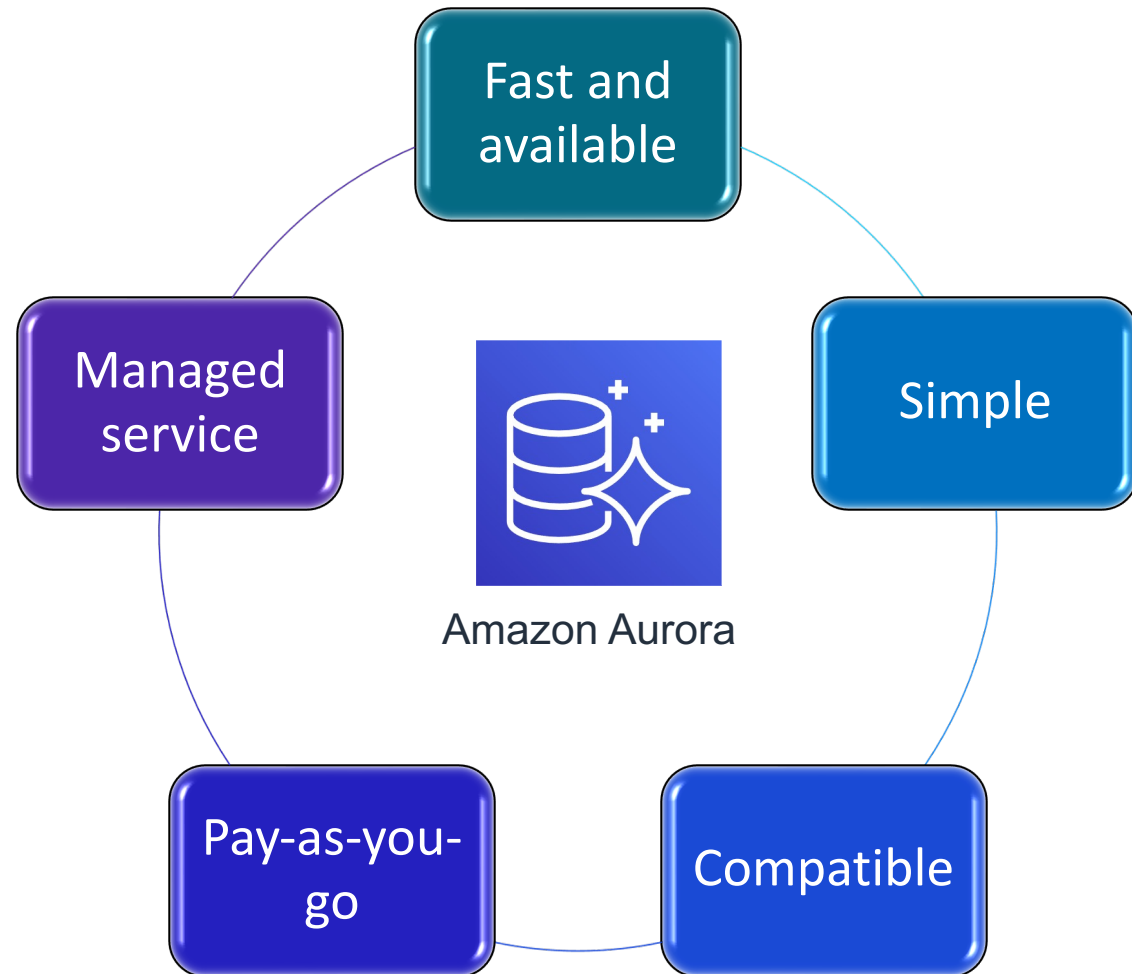


- Amazon RDS support multiple open source and commercial database engines.
- Amazon RDS instance class lets you choose the instance type (compute), while storage type helps you define the storage type, size and IOPS for the DB.
- RDS Standby and read replicas provide enhanced availability and durability for DB Instances.
- Automated backups and manual snapshots provide methods for backing up and restoring your DB.

Module 6: Computing (Database Services)

Amazon Aurora

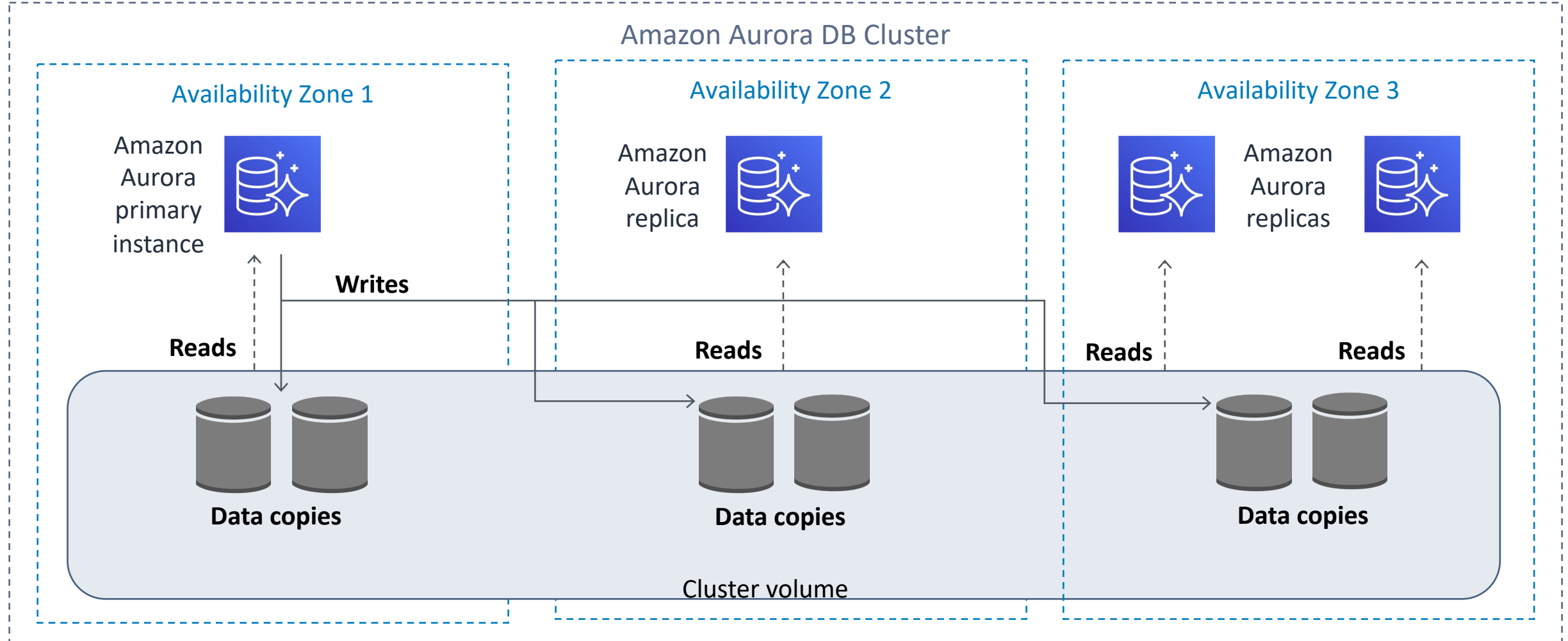
Amazon Aurora benefits



Benefits of **Amazon Aurora** include:

- A MySQL- and PostgreSQL-compatible relational database
- Combines performance and availability of high-end commercial databases
- Simple and cost-effective open-source database

Amazon Aurora high availability architecture



Module 6: Computing (Database Services)

Amazon DynamoDB

Amazon DynamoDB benefits

Amazon DynamoDB is a fast and highly scalable non-relational database service.

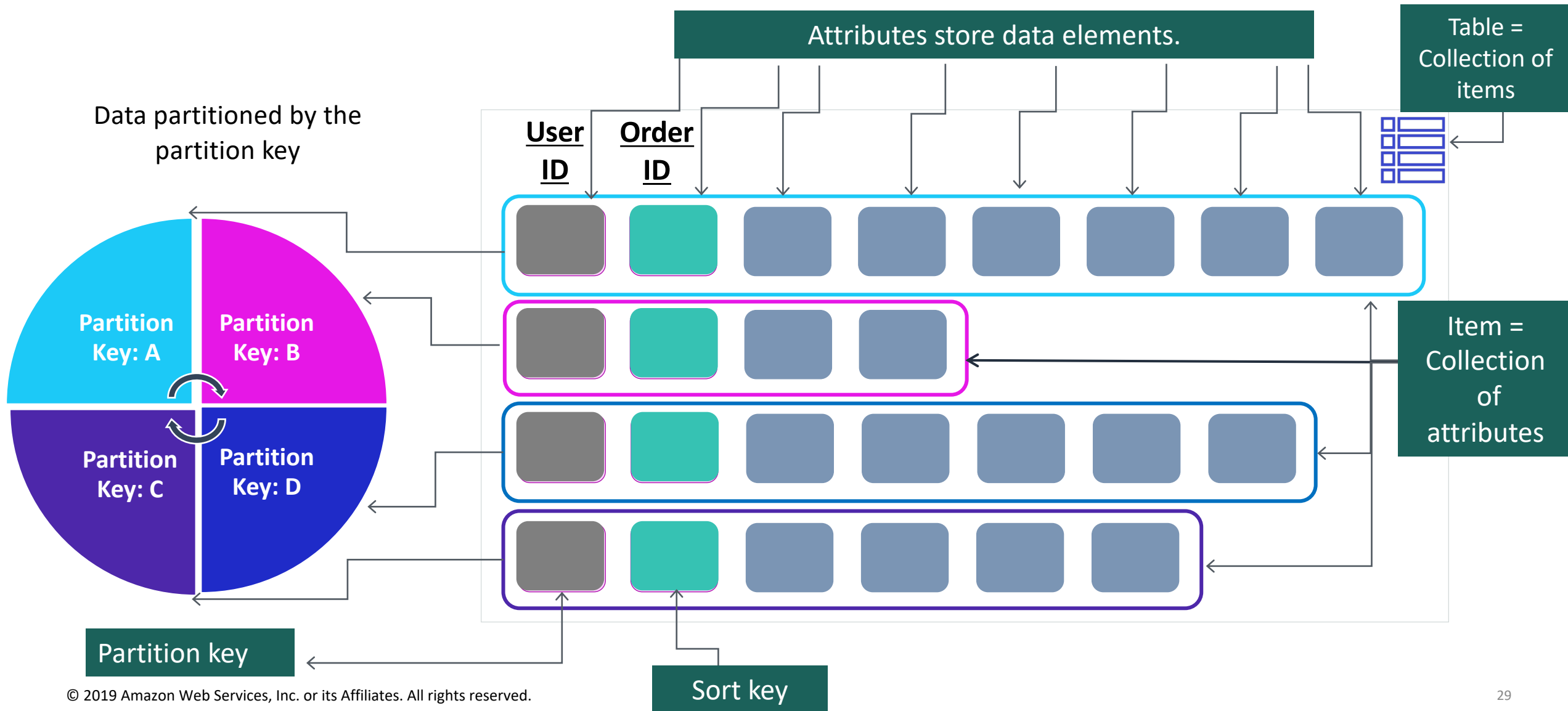
Benefits of **Amazon DynamoDB** include:

- Fully managed
- Low-latency queries
- Fine-grained access control
- Flexibility

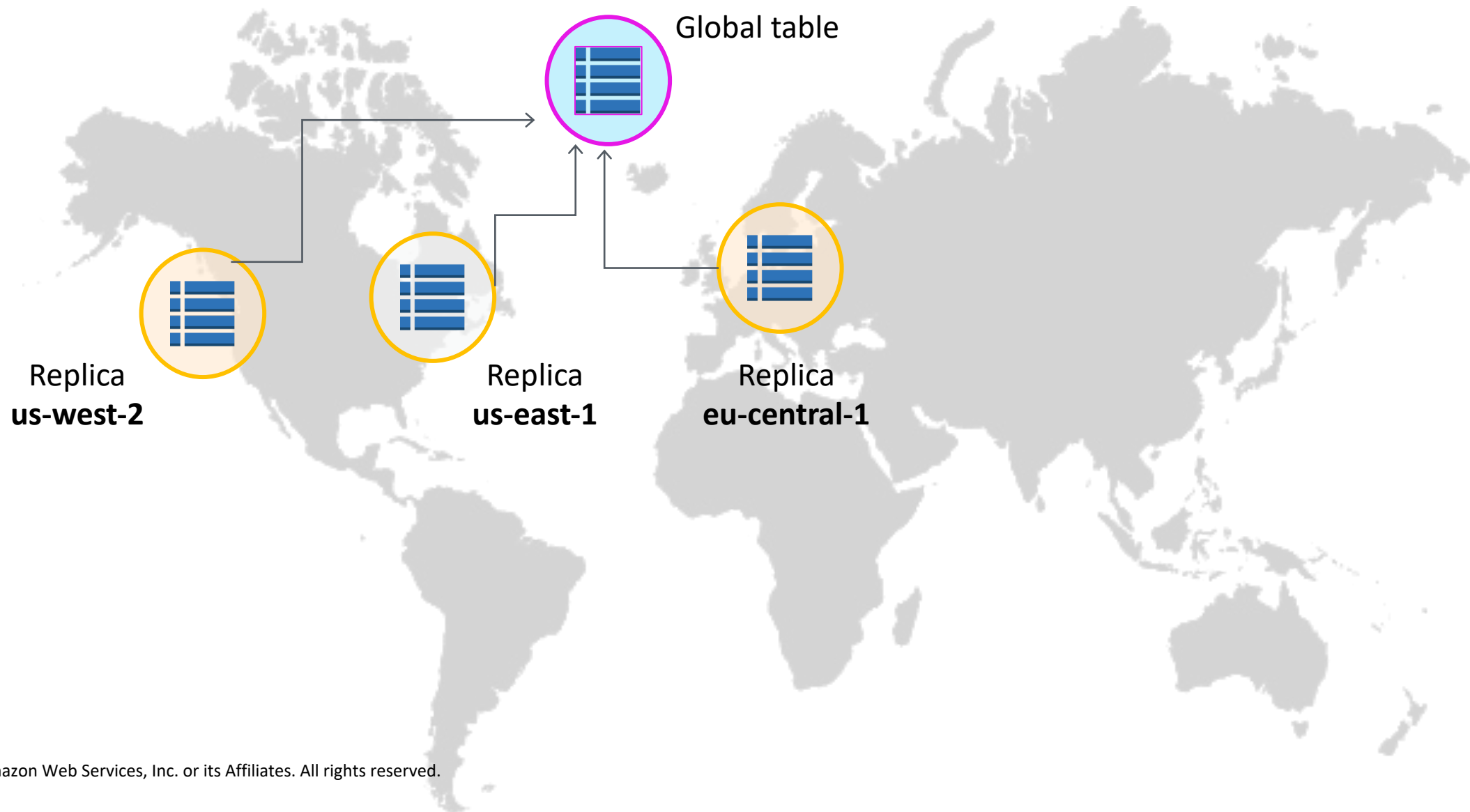


Amazon DynamoDB

Amazon DynamoDB data model



Amazon DynamoDB global tables



Key Takeaways



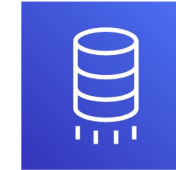
- Understand the key *data model* differences between SQL and NoSQL.
- Understand that the *scaling* and *performance* features of Amazon DynamoDB make it ideal for high-throughput, latency-sensitive applications.

Module 6: Computing (Database Services)

AWS Database Migration Service (AWS DMS)

Database Migration Service (AWS DMS)

- **AWS Database Migration Service (AWS DMS)**, along with the **AWS Schema Conversion Tool (AWS SCT)** enables you to migrate your database to AWS with minimal downtime.
- Supports most widely used commercial and open-source databases
- Keeps your applications running during the migration



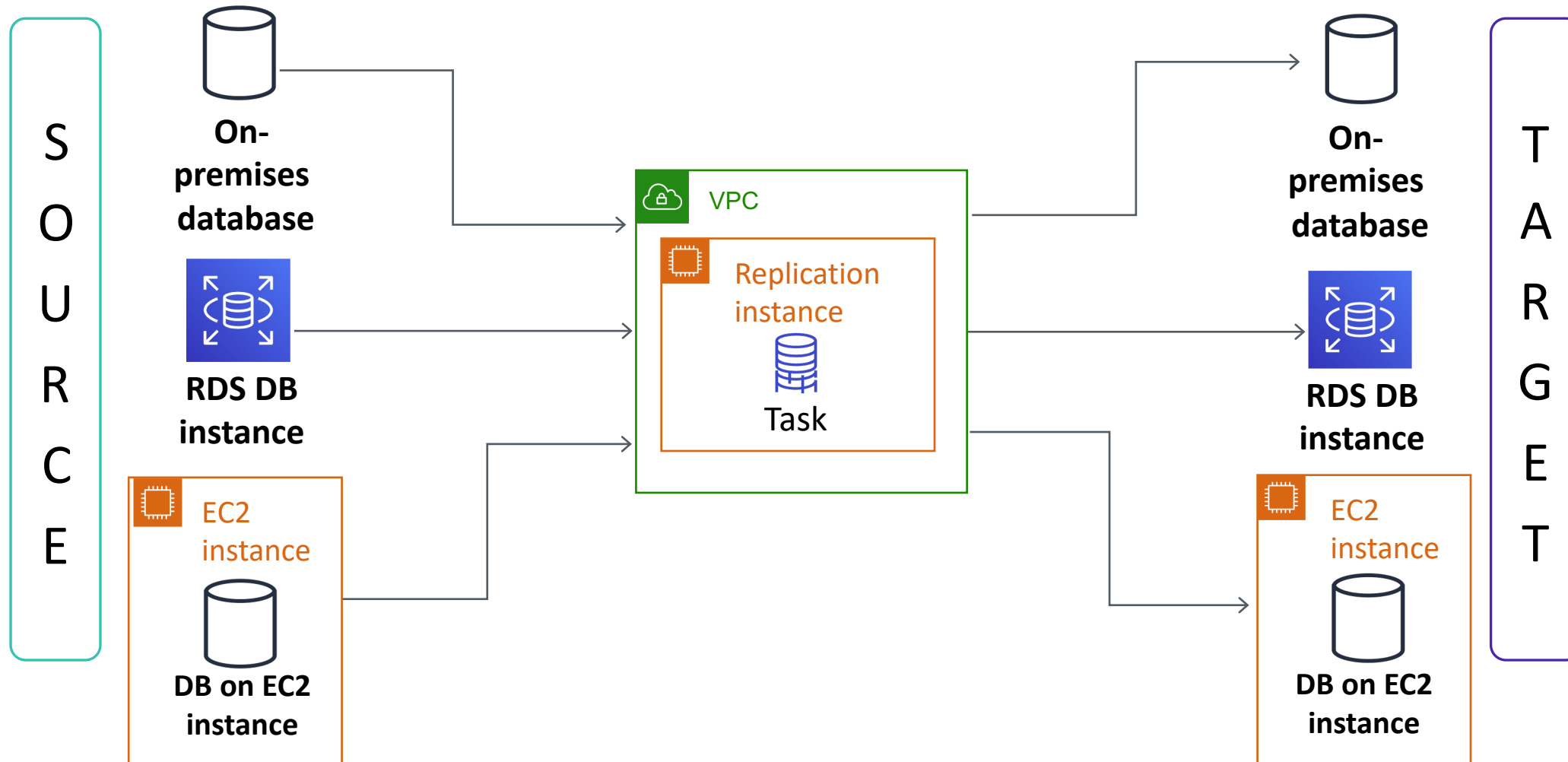
AWS Database Migration Service



DMS Source/Target Combinations

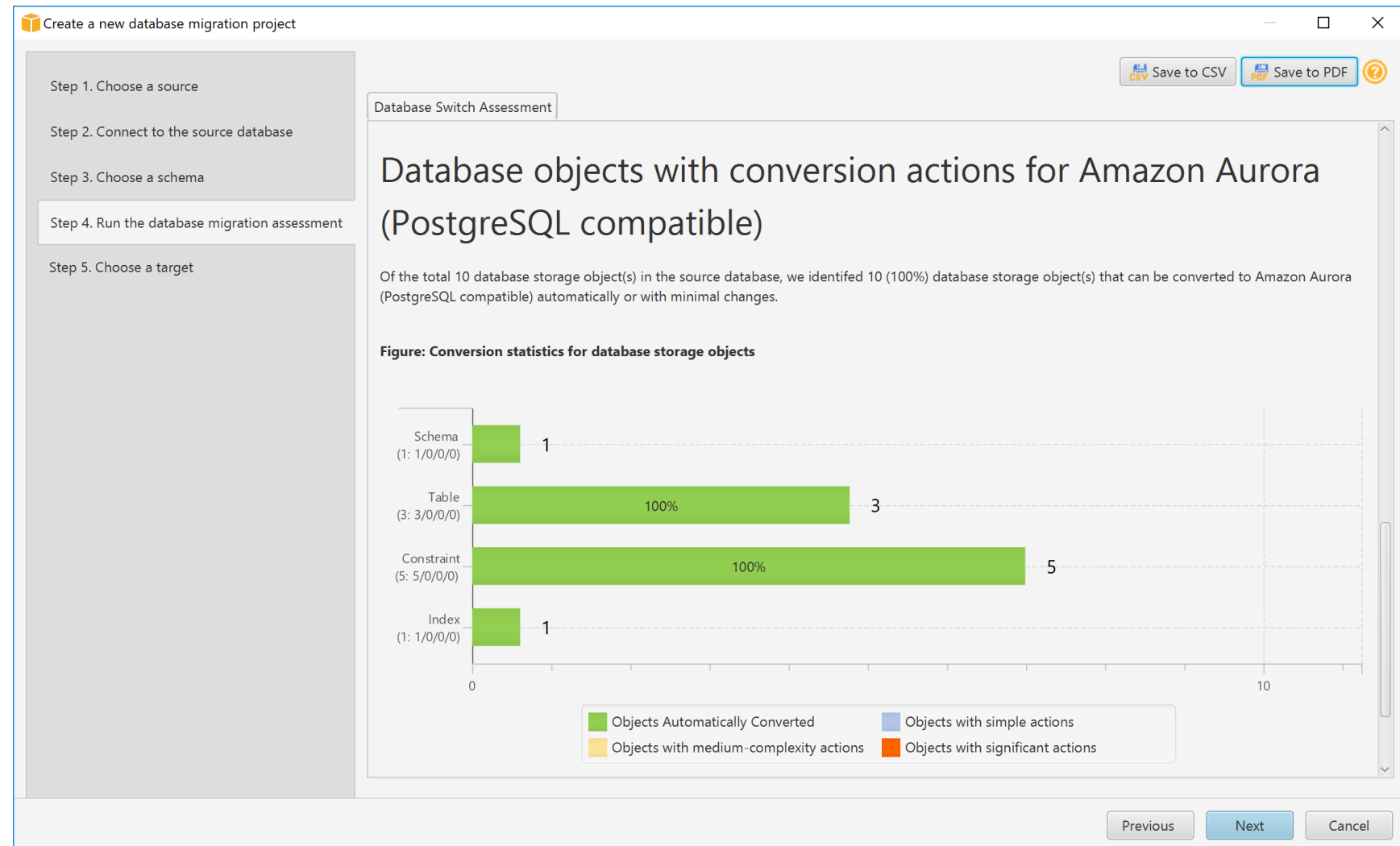
Source/Target	On Premises	Amazon RDS, Redshift, Amazon S3, and DynamoDB	Amazon EC2
On Premises	Not supported	Supported	Supported
Amazon RDS and Amazon S3	Supported	Supported	Supported
Amazon EC2	Supported	Supported	Supported
Azure SQL	Not supported	Supported	Supported

AWS DMS architecture



AWS Schema Conversion Tool

The **AWS Schema Conversion Tool** helps automate many database schema and code conversion tasks when migrating from source to target database engines.



Conversions supported by AWS SCT

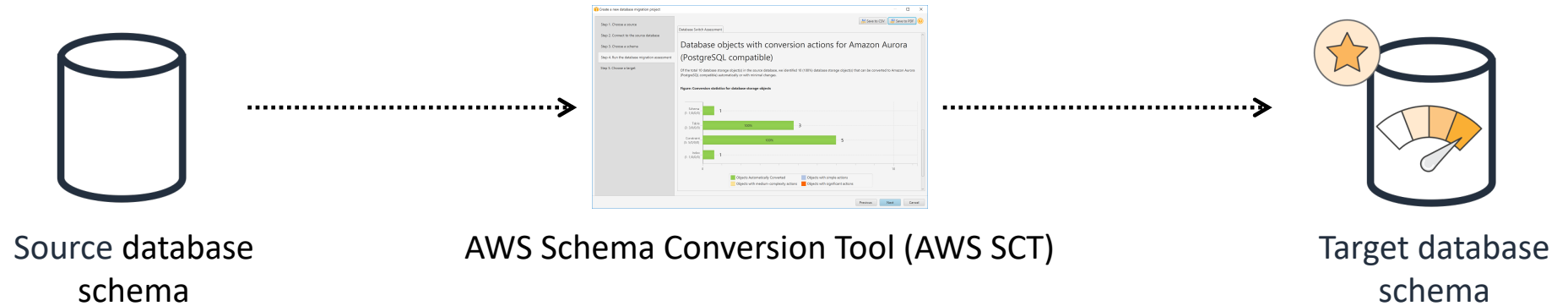
The **AWS Schema Conversion Tool** easily converts:

- Source database schema
- Views
- Stored procedures
- Functions

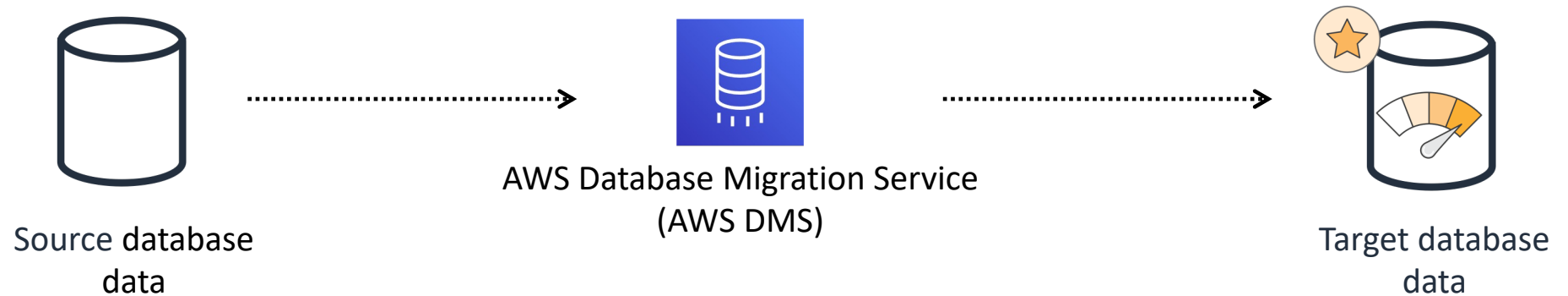
Source Database	Target Database on Amazon RDS
Oracle Database	Amazon Aurora, MySQL, PostgreSQL, MariaDB
Oracle Data Warehouse	Amazon Redshift
Microsoft SQL Server	Amazon Aurora, Amazon Redshift, MySQL, PostgreSQL, MariaDB
Teradata	Amazon Redshift
IBM Netezza	Amazon Redshift
Greenplum	Amazon Redshift
Vertica	Amazon Redshift
MySQL and MariaDB	PostgreSQL
PostgreSQL	Amazon Aurora, MySQL, MariaDB
Amazon Aurora	PostgreSQL
IBM Db2 LUW	Amazon Aurora, MySQL, PostgreSQL
Apache Cassandra	Amazon DynamoDB

Database migration process summary

Step 1



Step 2



Key Takeaways



- AWS DMS and AWS SCT help migrate homogenous and heterogeneous databases from on premises data centers and cloud instances to AWS.
- Plan migration steps based on availability requirements.

In summary, this module discussed:

- The types of managed database services that are offered by AWS and their recommended use.
- The main features and benefits of Amazon Relational Database Service (Amazon RDS), Amazon Aurora, and Amazon DynamoDB.
- The benefits of the AWS Database Migration Service (DMS) and the capabilities of the AWS Schema Conversion Tool (AWS SCT).

To finish this module:

Complete:  **Knowledge check**

Test Strategies



Module 6 Scenario:

A FinTech bank has created a new credit card line; therefore, will need to expand its database to capture new customer profiles and link existing customer profiles to the new card line, while maintaining security and PCI compliance requirements. The bank is entirely in the cloud and provides their customers with unique financial visualizations about their financial habits. Their customers use a mobile app and website for their account transactions and profiles.

What AWS service should be prioritized for the bank's database if they want to graph connected data on their customers' financial behaviors?



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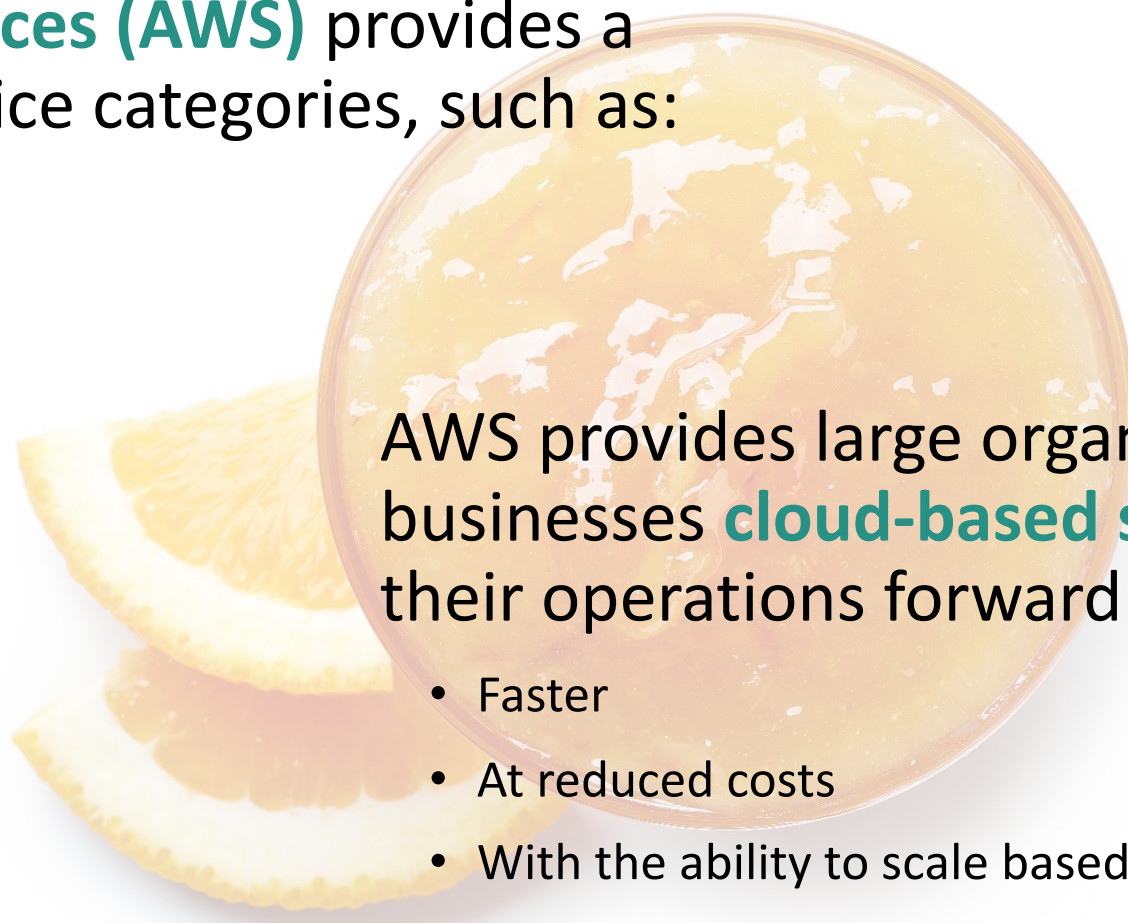
- A. Amazon Aurora
- B. Amazon DynamoDB
- C. Amazon ElastiCache
- D. Amazon Neptune
- E. Amazon Redshift

Systems Operations Project Scenario

Mom & Pop Café & Bakery

Amazon Web Services (AWS) provides a broad array of service categories, such as:

- Compute
- Storage
- Database
- Analytics

A decorative image featuring a glass bowl filled with orange juice and several orange slices floating in it, positioned behind the text on the right side of the slide.

AWS provides large organizations and small businesses **cloud-based solutions** to move their operations forward:

- Faster
- At reduced costs
- With the ability to scale based on demand

On completion of this project, you will be able to:

- **Experience** the communication challenges faced when attempting to apply technology as the solution to business problems.
- **Translate** customer requirements into a proposed technical solution.
- **Present** the proposed solution to the customer.
- **Use** AWS Best Practices for a frugal, resilient, and secure solution.



Customer Meeting Preparation

Mom & Pop Café & Bakery

Database Migration Request

Business Need: Managed Database

Emma is concerned about:

- Business data tracked in a spreadsheet on a local PC, which must be with her at all times.
- Potential to lose financial, point of sale (POS), and labor data.
- Spreadsheet is not secured, and is becoming challenging to use.
- No way to know when ingredient supplies are getting low.
- If the EC2 instance that hosts the order history database fails, they will lose all their data.



Sophie's Database Request



Sophie calls Alejandro about creating a database. She suggests a solution that aligns with **AWS Best Practices** within the **Well Architected Framework**.

However, Alejandro points out larger concerns:

- All the café's order data is on one EC2 instance
- This creates a potential for failure, which could shut them down

Alejandro suggests the order history should be **migrated to Amazon Relational Database Service (Amazon RDS)**.

Supply chain contact information should also be imported from the Emma's laptop to Amazon RDS.

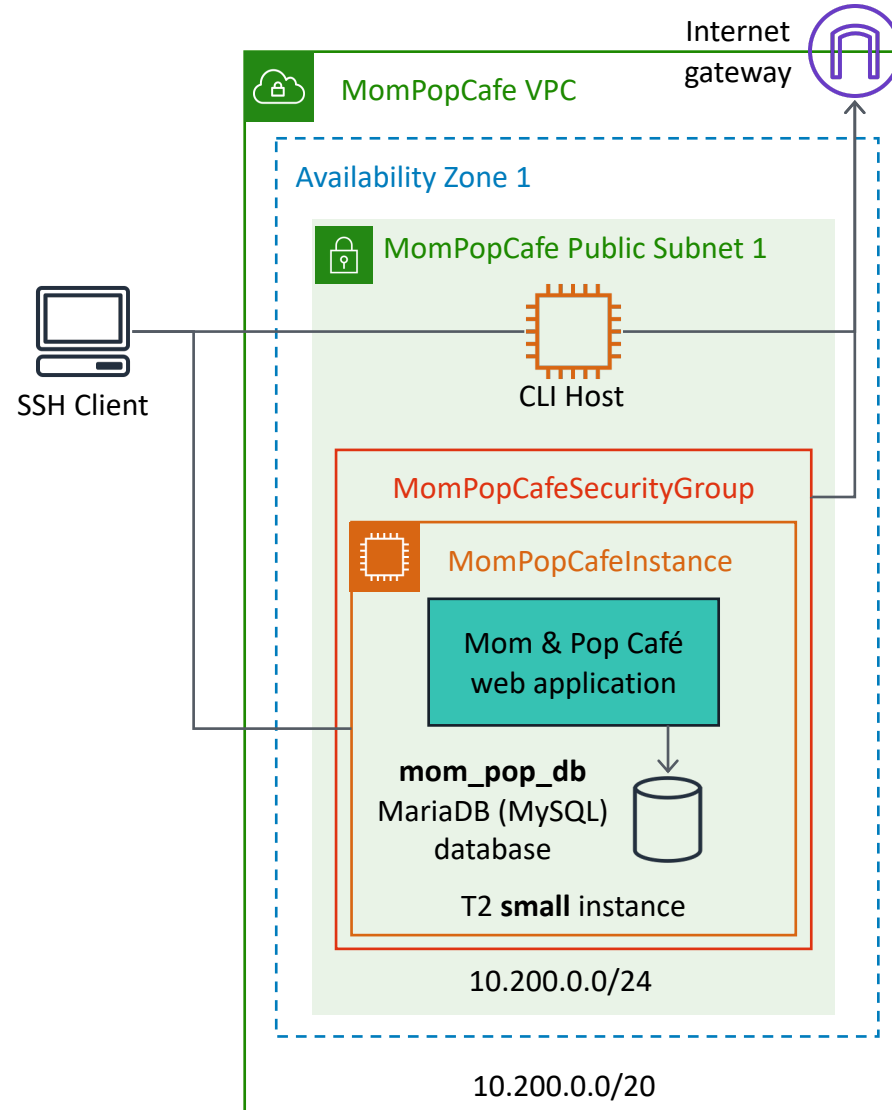
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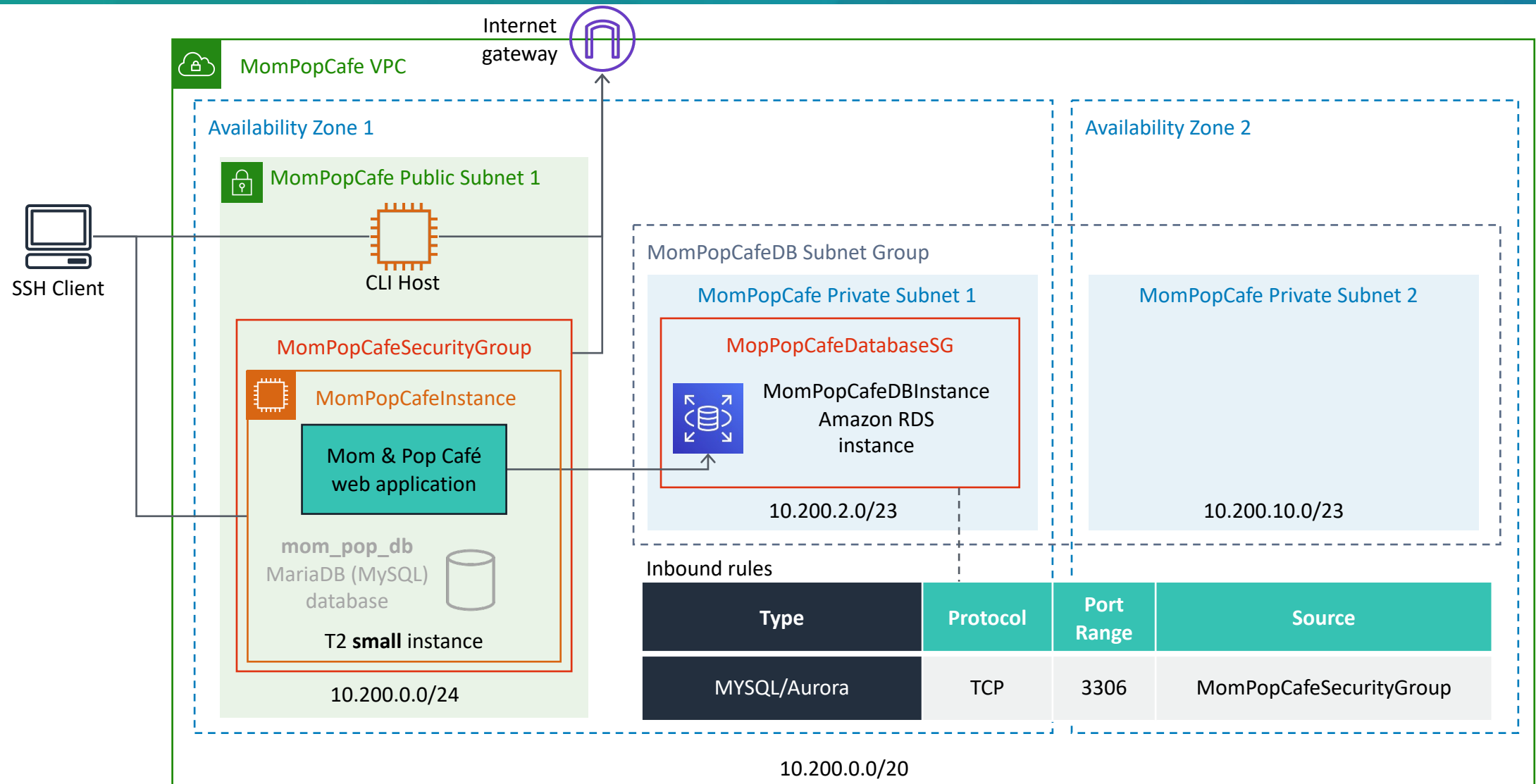


Learn more with
the Data Migration
Playbook

Activity 6: Migrate to Amazon RDS - Before migration topology



Activity 6: Migrate to Amazon RDS - After migration topology





Begin Activity #6: Migrate Mom & Pop Café from MySQL on an Instance to Amazon RDS

Benefits of Amazon RDS

What are six benefits of using Amazon RDS?

Reason #1

Reason #2

Reason #3

Reason #3

Reason #5

Reason #6

Benefits of Amazon RDS

What are six benefits of using Amazon RDS?

Reason #1

Easy to
administer

Reason #2

Highly scalable

Reason #3

Available and
durable

Reason #3

Fast

Reason #5

Secure

Reason #6

Inexpensive

Project time: Troubleshooting Knowledge Base



Thank you

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